Glacier upward { sand } do later }

GS FOUNDATION 2024-25

GEOGRAPHY - 19

Lecture 21

Landforms and Their Evolution

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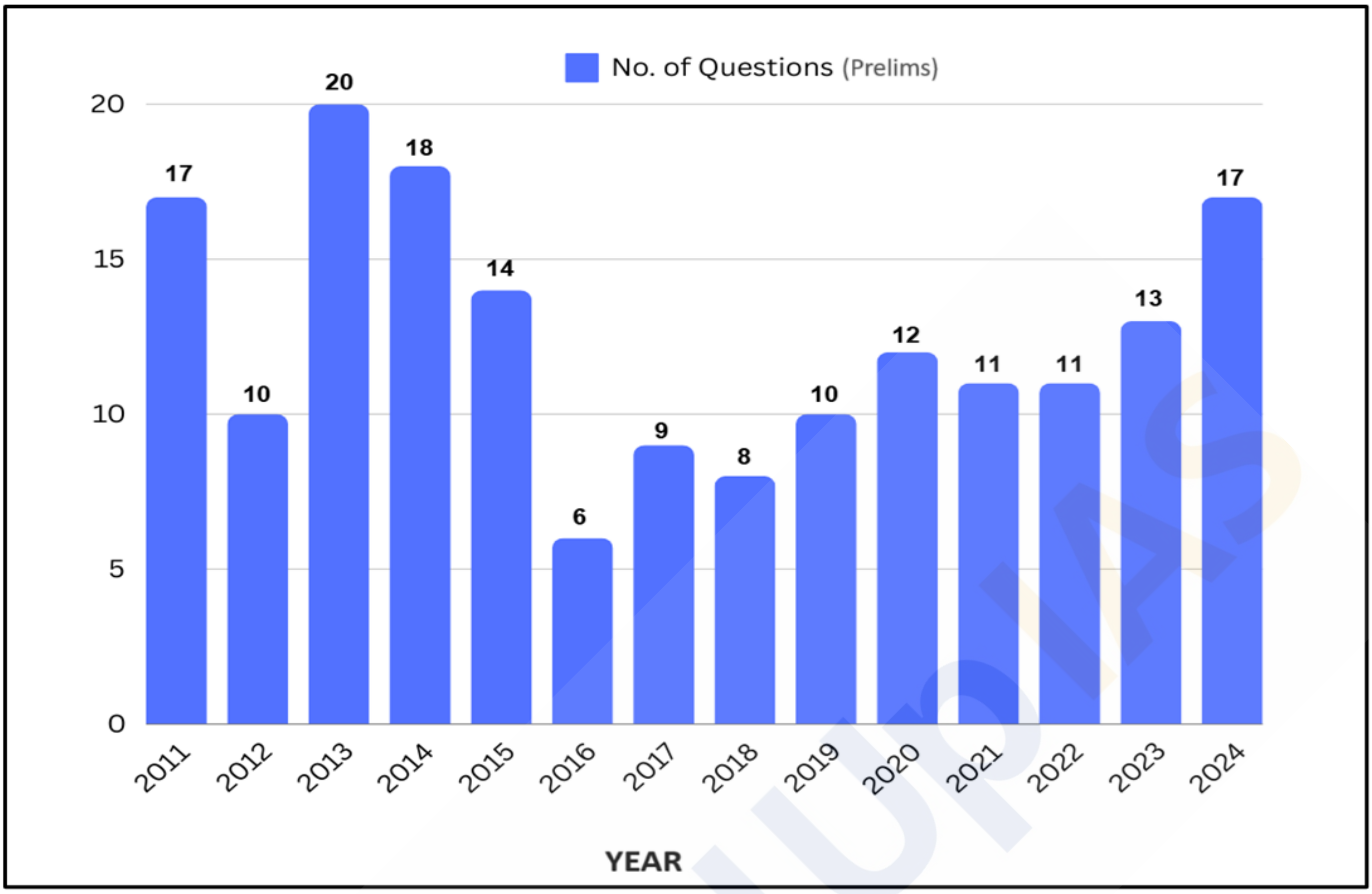

Topic-wise Trend Analysis (2011 – 2024)

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Landforms and Their Evolution

Introduction:

After weathering processes act on the materials that make up the earth's surface, geomorphic agents such as **running water, groundwater, wind, glaciers,** and **waves** take over to perform erosion. It is well known that erosion causes significant changes to the earth's surface. **Following erosion, deposition occurs, which also leads to changes on the surface of the earth.**

Landforms are small to medium-sized areas of the earth's surface. When several related landforms come together, they create landscapes, which are larger tracts of the earth's surface. Each landform has a **distinct physical shape, size, and composition,** and these characteristics result from specific geomorphic processes and agents. Most of these processes and agents work slowly, so it takes a long time for their effects to become noticeable. Every landform has an origin and evolves over time.

Once a landform is created, it can **change in shape, size, and nature either gradually or rapidly,** depending on the continued action of geomorphic processes and agents. Changes in climatic conditions and the vertical or horizontal movements of land masses can alter the intensity or nature of these processes, leading to further modifications in landforms.

Evolution in this context refers to the **stages of transformation** that parts of the earth's surface undergo, either changing from one landform to another or altering individual landforms after their initial formation. Each landform has a history of development and changes over time, similar to the stages of life—**youth, maturity,** and **old age.**

Landforms Made by Running Water or Fluvial Landforms:

Fluvial landforms are **those landforms which are shaped and modified by the running water.** The term "fluvial" is derived from the Latin word **"fluvius,"** meaning **'river.'** Running water has sculpted most of the land surface across the world in comparison to other agents of erosion like wind or Glacier. Fluvial landforms exhibit a wide range of dimensions, from small features like rills to large-scale units such as major rivers and their extensive drainage basins. Remarkably, **rivers flowing to the oceans drain about 68% of the Earth's land surface.**

Rivers typically originate in upland regions where the slope facilitates runoff. These upland areas serve as the catchment zones for rivers, with the **crest of mountains forming divides or watersheds.** From these divides, streams flow down the slopes. The initial stream that forms due to the slope is known as a **'consequent stream.'** As this stream erodes the surface, it is joined by numerous tributaries from either side.

A **drainage basin,** or **watershed,** is a **fundamental landscape unit in fluvial geomorphology.** It includes a primary, or trunk, river and its tributaries. The drainage basin is crucial for understanding how water and sediments move through a landscape.

Erosion → At start Deposition
↳ At End

Erosion by Running Water occurs through various processes:

- **Hydration -** The **physical force of running** water wearing down rocks.
- **Corrosion or Solution - Chemical weathering** caused by the interaction of water with minerals in rocks.
- **Attrition -** When the **stream loads** (pebbles, sand, silt etc) **move together, they cause their own wear and tear** by colliding with one another. This rounding and shaping of these pebbles and boulders amongst themselves is called attrition.
- **Corrasion or Abrasion - Solid particles** carried by the river **strike against rock surfaces,** wearing them down. → Scrapping the surface
- **Downcutting (Vertical Erosion) -** The **deepening of a stream** channel. → Generally at Mountain top V. shaped formation.

River capture is generally through headward erosion → Eg. Ganga & yamuna River

- **Lateral Erosion - The widening of a stream** channel by eroding its banks. → In general Plain area
- **Headward Erosion -** Erosion at the source of a stream, **extending the stream channel upstream.** → Top of head ↳ sources can expand. Eg. Ganga
- **Hydraulic Action (Pressure) -** The **mechanical loosening and removal of material** by the force of river water.
- **Braiding -** Braiding occurs when the **main river channel splits into multiple, narrower channels.** This results in a **network of river channels separated by small, often temporary, islands** called **braid bars.** Braided rivers are common in areas with a low slope and/or a high sediment load. ↳ Majuli Island, Brahmaputra

Erosional Landforms by Running Water:

Valleys:

Valleys represent a major erosional landform, evolving through various stages over time. The initial stage of valley formation begins with **small and narrow rills,** which are **tiny channels** carved into the landscape by flowing water. As these rills are subjected to **continuous erosion, they gradually expand into larger and wider channels** known as **gullies.** With ongoing erosion, **gullies further deepen, widen,** and **lengthen,** eventually transforming into **valleys.**

Types of Valleys:

- **V-Shaped Valleys -** As the name suggests, these valleys have a **pronounced V-shape** when viewed in cross-section. They are typically formed in the **upper course of a river** where the flow of water has a high velocity, cutting sharply into the landscape.

V → Shaped of Erosion

- **Gorge -** A gorge is a type of valley that is **extremely deep with very steep to nearly vertical sides.** Unlike other valleys, a gorge maintains a relatively **consistent width from its top to its bottom.** Gorges typically form in **areas where the bedrock is very hard and resistant to erosion.**

- **Canyon -** A canyon is similar to a gorge but has distinct characteristics. It features **steep, step-like side slopes** and is generally as deep as a gorge. However, a canyon is **wider at its top than at its bottom.** Essentially, a canyon is a variant of a gorge but is commonly formed in **areas with horizontally bedded sedimentary rocks.** Eg. Grand Canyon

Eg. Gandikota canyon → Pennar River → Eastern Ghat

Potholes and Plunge Pools:

- **Potholes -** They are **circular depressions found in riverbeds,** created by the erosive action of flowing water combined with the **abrasion caused by rock fragments.** When a small depression forms, pebbles and boulders get trapped and rotated by the water, gradually enlarging the depression.

- **Plunge Pools -**They are **large potholes that are significantly deeper and wider.** They form at the base of waterfalls due to the intense impact of falling water and the swirling motion of boulders. Plunge pools play a crucial role in the **deepening of valleys by continually eroding the bedrock.**

Waterfalls and Rapids:

- **Waterfalls -** It occur where a **river makes a steep descent over rocky slopes or ledges.** As the river plunges down with great velocity, it **causes significant erosion,** particularly at the base of the waterfall, where **plunge pools** are formed. These waterfalls often occur where the **river flows from softer rock, such as limestone or sandstone, to harder rock like granite.** The softer rock erodes more easily, creating a steep drop over the more resistant rock. Waterfalls, also known as cascades, are spectacular features found in various locations around the world. **Example - Niagara Falls** at the USA-Canada border, **Jog Falls** in India, and **Victoria Falls** in Zambia.

- **Rapids -** It form in areas of shallow, **fast-flowing water,** typically in **younger streams.** They are characterized by turbulent water and a series of small waterfalls or cascades within the stream. These features are common in river channels where the **water flows over a rocky bed, creating a series of drops and waves.** Rapids are **popular sites for adventure sports** like white-water rafting, with notable locations such as Rishikesh in India offering thrilling experiences for enthusiasts.

These are due to alternate soft & hard rock bed. Soft rock erode fast than hard rock.

snake like structure

Incised or Entrenched Meanders:

↳ Generally occurs in flood plain where lateral erosion is

Special type Meander is Incised Meander

- A **meander** is defined as a **pronounced curve or loop** in the course of a river channel. **Incised or entrenched meanders** are deeply embedded meanders **found in hard rock formations.** They occur where rivers experience **rapid vertical erosion, cutting deeply into the riverbed without significantly eroding the lateral sides.**

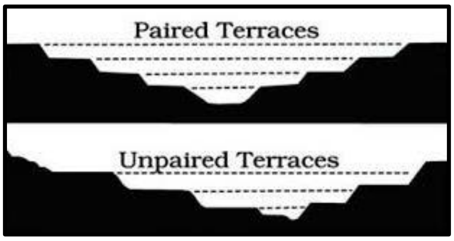
- As the river continues to cut into the bedrock, these loops become deeper and wider, eventually forming deep **gorges** or **canyons.** Meandering courses are typically found over floodplains and delta plains where the stream gradients are very gentle, promoting lateral erosion and creating wide, looping bends.
- **Oxbow Lake -** Sometimes, because of intensive erosion action, the **outer curve of a meander** gets accentuated to such an extent that the inner ends of the loop come close enough to get **disconnected from the main channel and exist as independent water bodies** called as **'oxbow lakes.'** These water bodies are converted into swamps in due course of time. In the Indo-Gangetic plains, southwards shifting of Ganga has left many oxbow lakes to the north of the present course of the Ganga.

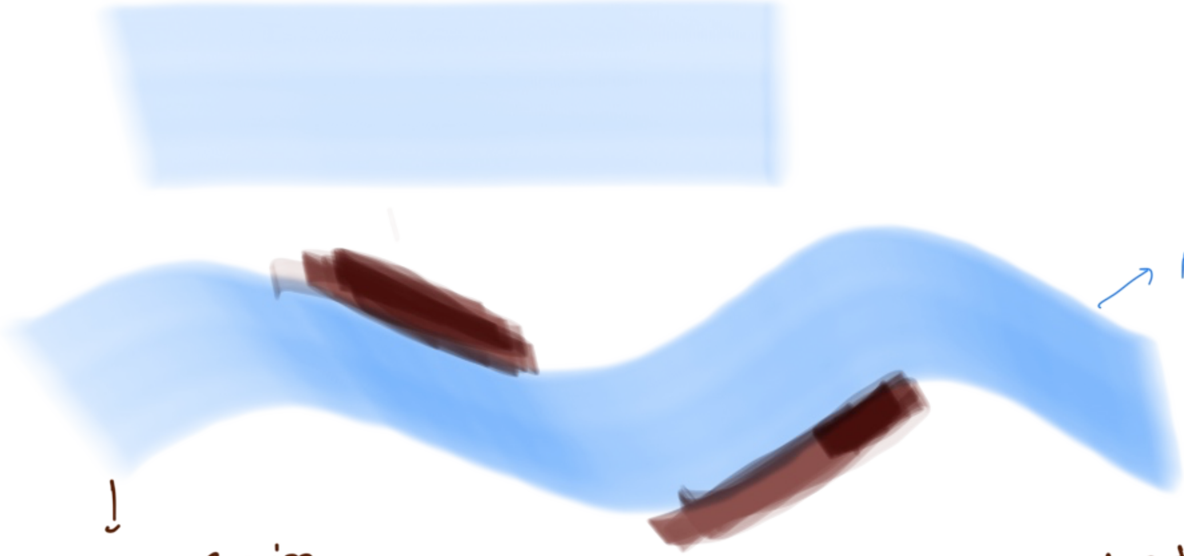
eg. Kanwar lake → river Gandak

River Terraces:

- River terraces are **flat surfaces** that mark former levels of a **river's valley floor or floodplain.** They are primarily erosional features, **created by the vertical erosion of a river cutting down into its own previous floodplain deposits.** River terraces can be found at various elevations along a river valley.

- **When terraces are at the same elevation on both sides** of the river, they are known as **paired terraces.** In contrast, **unpaired terraces occur when terraces are present on only one side** of the river or at different elevations on opposite sides.



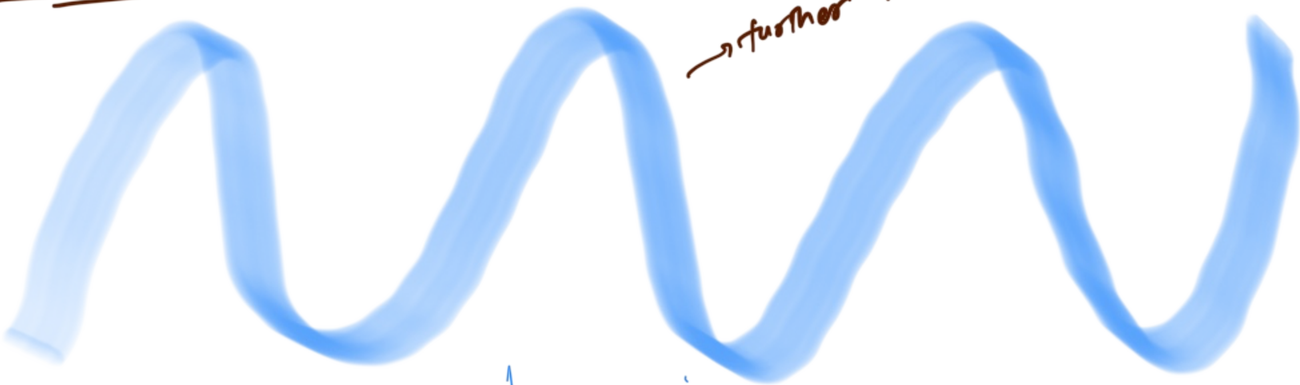


Manders & sedimed deposition

outbank Erosion
other side deposition

further Meandar

due to deposition at one side than other



overtime

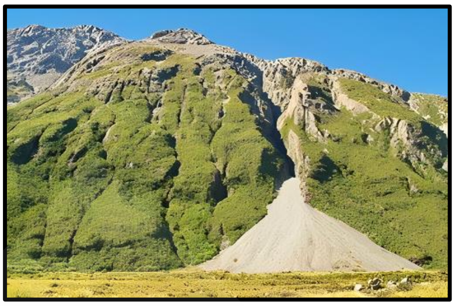
Sediment deposit

This part seperate

This is oxboew lake

Depositional Landforms by Running Water:

Alluvial Fans and Cones:



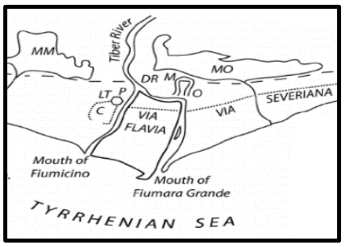
- Alluvial fans are distinctive, **fan-shaped deposits of sediment** that form where a high-gradient stream loses energy as it flows onto a flatter plain, typically at the **base of mountain slopes.** These formations occur when **streams carrying coarse sediment descend steep mountain slopes and suddenly encounter the lower gradient of a foot slope plain.** The abrupt decrease in slope reduces the stream's carrying capacity, causing it to deposit the heavier sediment it can no longer transport. This process results in a broad, **cone-shaped accumulation of coarse materials,** creating what is known as an **alluvial fan.**
- In **humid regions, alluvial fans usually have low cones with gentle slopes,** while in **arid and semi-arid climates,** they appear as **high cones with steep slopes.** The streams on these fans frequently shift their positions, forming multiple channels called **distributaries.**

Deltas: → Generally triangle based shape

- Deltas are **sedimentary deposits** that form at the **mouths of rivers where they flow into a standing body of water, such as an ocean, sea, or lake.** Similar to alluvial fans in terms of their depositional nature, deltas develop in a different environmental setting. As rivers carry sediment towards their mouths, the **reduction in flow velocity causes sediment to settle and accumulate, forming a delta.** Unlike alluvial fans, delta deposits are typically **well-sorted and exhibit clear stratification** due to the continuous action of waves, tides, and currents that redistribute the sediments. Deltas often have a **variety of sub-environments,** including **distributary channels, levees,** and **floodplains,** each contributing to the complexity of the deltaic system.
- **Types of Deltas:**
  - **Arcuate Delta (Fan-shaped Delta) -** Arcuate deltas are **triangular or fan-shaped** with a convex seaward margin. They are formed in **regions with a high sediment load and low wave** and tidal energy, which allows the river to spread out and deposit sediments evenly.

    - A prime example of an arcuate delta is the **Nile Delta in Egypt.**
  - **Bird's Foot Delta -** Bird's foot deltas resemble the **claws of a bird's foot,** with several distributaries **extending out into the sea.** These deltas develop where **river-dominated processes prevail over wave and tidal actions,** allowing distributaries to extend far out into the water body.

    - **The Mississippi River Delta in the USA** is a notable example.
  - **Cuspate Delta -** Cuspate deltas have a **tooth-like or cusp-shaped structure** with a pointed seaward margin. They form where **strong wave action redistributes sediments** on either side of the river mouth, shaping the delta into a pointed structure.



    - The **Tiber River Delta** in Italy is an example of a cuspate delta.
  - **Estuarine Delta -** Estuarine deltas form **within the estuary of a river.** They develop where a river deposits sediment into a long, narrow estuary, and tidal processes significantly shape the delta.

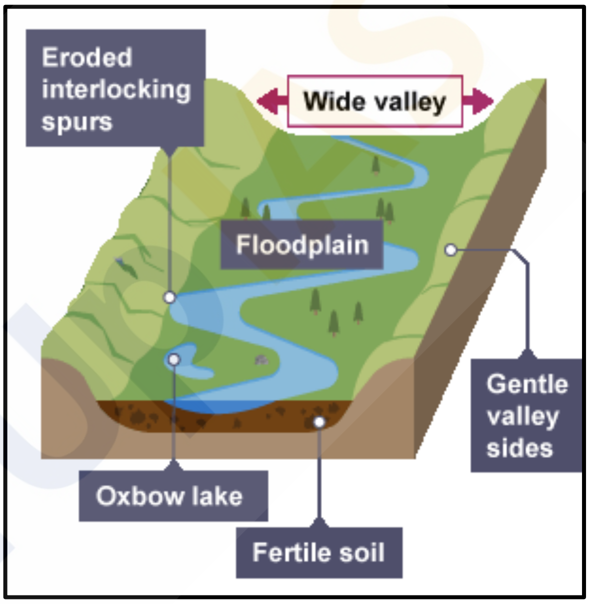
    - **The Seine River Delta** in France is a typical example of an estuarine delta.

Normade River in Arabian sea form Aliabet Estuary

- <u>**Conditions Favourable for Delta Formation**</u>:
    - **Active Erosion -** Extensive vertical and lateral erosion in the upper course of the river provides sediments for delta formation.
    - **Sheltered Coast -** The coast should be sheltered and preferably tideless to allow sediment accumulation.
    - **Shallow Sea -** The adjoining sea should be shallow to prevent sediment dispersion in deep waters.
    - **No Large Lakes -** The river should not have large lakes that filter out sediments.
    - **No Strong Currents -** Strong currents running at right angles to the river mouth should be absent to avoid washing away sediments.

<u>Floodplains</u>:

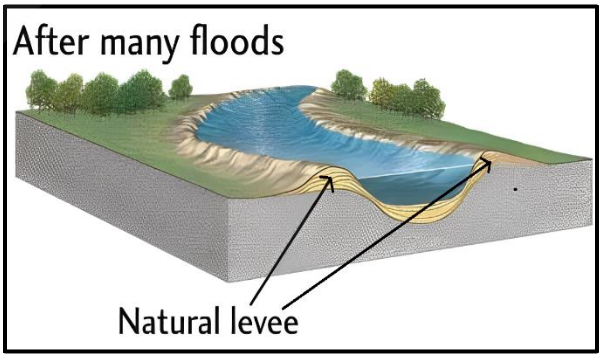
- Floodplains are significant landforms created by river deposition. When a **stream flows from a steeper to a gentler slope, it loses energy and deposits large-sized materials first.** As the water slows down further, it carries finer materials like sand, silt, and clay, **depositing them over the riverbed and spilling over the banks during floods.**

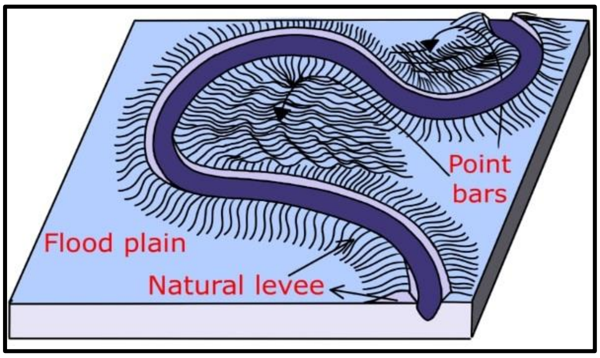


- **The floodplain above the bank,** known as the **inactive floodplain,** contains **two main types of deposits –**
    - **Flood deposits**
    - **Channel deposits.**
- In these areas, **river channels often shift laterally,** changing their courses and leaving behind cut-off channels. These abandoned channels gradually fill with coarse deposits, contributing to the floodplain's structure.
- Floodwaters typically **carry finer materials, such as silt and clay,** which are deposited across the floodplain. When these floodplains are part of a delta, they are specifically referred to as **delta plains.**

<u>Natural Levees and Point Bars</u>:

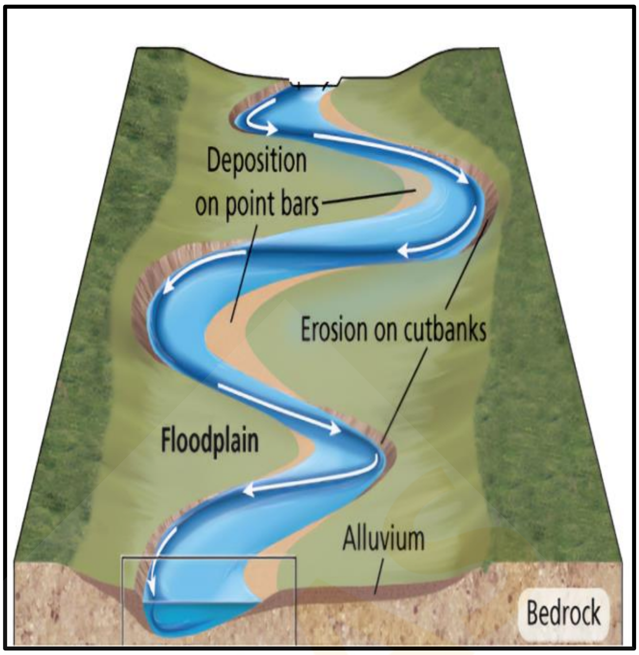
- **Natural levees -** These are the prominent features **found along the banks of large rivers.** These landforms are characterized by **low, linear ridges of coarse deposits that run parallel to the riverbanks.** During periods of flooding, rivers overflow their banks, and the **heaviest sediments are deposited closest to the river, gradually building up these ridges.** Over time, the levees may be cut into individual mounds due to erosional processes, but they typically remain as continuous barriers that help contain the river's flow.





- **Point bars -** Also known as **meander bars,** are formed on the **inner, concave side of river meanders.** As the river flows around bends, it erodes the outer banks and deposits sediments on the inner banks. These deposits **accumulate in a linear fashion, creating point bars that are almost uniform in profile and width.** The sediments within point bars vary in size, including a mixture of fine and coarse materials. **Point bars are integral to the meandering process** of rivers, contributing to the dynamic reshaping of the river channel and floodplain over time.

Meanders:



- In large floodplains and delta plains, rivers often exhibit **loop-like patterns known as meanders.** Meander is **not a landform** but is only a **type of channel pattern.**
- When water flows over areas with very gentle slopes, it tends to work laterally rather than vertically. This **lateral movement exerts pressure on the banks of the river, causing erosion.** The force of the water pushes against the banks, carving out curves and bends, leading to the formation of **meanders.**
- The banks of rivers with meandering channels are often composed of **unconsolidated alluvial deposits.** These materials are loose and easily eroded, making them susceptible to the lateral pressure exerted by flowing water. Any irregularities or weaknesses in these deposits are exploited by the river, contributing to the development of meanders.
- **The Coriolis force,** which affects the movement of fluids on a rotating Earth, also **plays a role in the formation of meanders.** This force causes the water to be deflected as it flows, similar to how it influences wind patterns. The deflection enhances the lateral movement of the water, promoting the meandering pattern.
- The tendency for a river to meander is influenced by the **balance between erosion and deposition.** If there is no significant deposition or erosion, the river is less likely to develop pronounced meanders. **Erosion on the outer banks of bends and deposition on the inner banks** (creating point bars) are crucial processes that sustain the meandering pattern.

Landforms Made by Groundwater:

Groundwater acts as a significant erosional force, capable of dissolving solid rock over time. It performs **erosion, transportation,** and **deposition,** leading to the creation of several picturesque topographical features. This process begins when **rainwater absorbs carbon dioxide (CO_2)** as it falls through the atmosphere. The absorbed CO_2 combines with the water to form a **weak carbonic acid solution.** When this slightly acidic water infiltrates the ground**, it percolates through the pore spaces in the soil and the cracks and fractures in the underlying rock.** This subsurface water movement is what we refer to as groundwater.

One of the most notable effects of groundwater is its ability to **dissolve limestone.** Limestone, composed mainly of **calcium carbonate,** reacts readily with carbonic acid. As groundwater flows through limestone formations, it gradually dissolves the rock, creating various landforms characteristic of regions known as **'Karst topography.'**

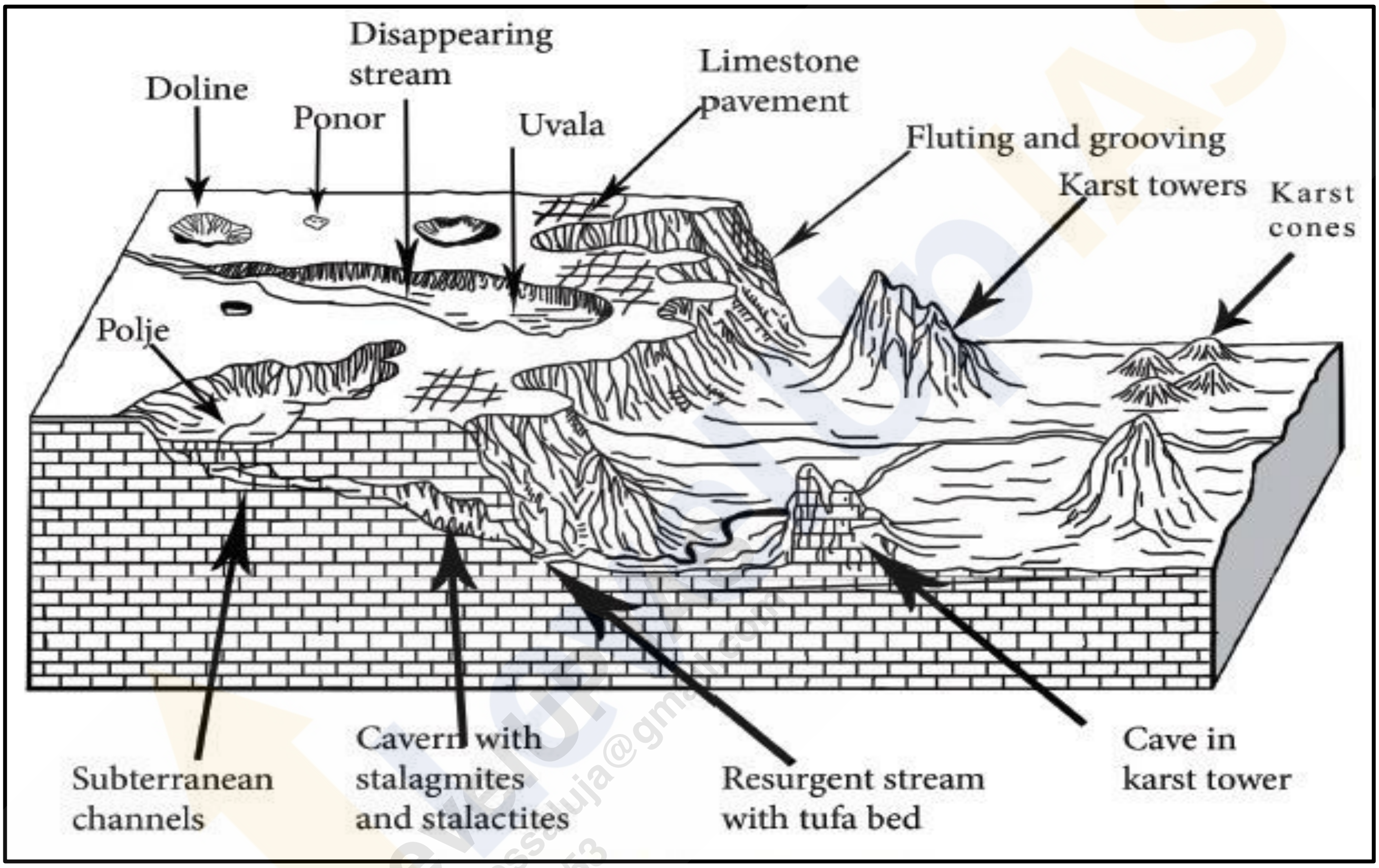
Karst Topography:

The term **"Karst"** is derived from the **Karst region** along the **Adriatic Sea coast in Croatia** (formerly Yugoslavia), where such formations are prominent. Karst topography is identified in areas where the landscape is shaped predominantly by the **dissolution of soluble rocks, primarily limestone and dolomite.**

Karst landscapes are found in various regions around the world, including the **Causses of France, the Kwangsi area of China, the Yucatán Peninsula in Mexico,** and several locations in the **United States** such as the **Midwest, Kentucky,** and **Florida.**

In India, karst topography is present in the **Vindhya region** (particularly southwestern Bihar), **parts of the Himalayas** (including Jammu & Kashmir, Robert Cave, Sahasradhara, the eastern Himalayas, and areas near Dehradun), **Pachmarhi** in Madhya Pradesh, **Gupt Godavari Cave** in Chitrakoot (Uttar Pradesh), the coastal areas near **Vishakhapatnam** (**Borra Caves**), and **Bastar** in Chhattisgarh and **Borra Caves** in Andhra Pradesh.

The action of groundwater through the processes of **solution** (dissolution of rock) and **deposition** (precipitation of minerals from groundwater) leads to the formation of unique landforms in these regions. These landforms can be broadly classified into **erosional and depositional types.**



Erosional Landforms by Groundwater:

Swallow Holes:

- Small to medium-sized **round to sub-rounded shallow depressions,** known as swallow holes, form on the surface of limestone through the process of solution. These features are common in limestone and karst areas.

Sinkholes (Doline):

- Sinkholes are prevalent in limestone or karst regions. They are openings that are more or less **circular at the top and funnel-shaped towards the bottom,** with sizes ranging from a few square meters to a hectare, and depths varying from less than half a meter to over thirty meters.

- Sinkholes can form solely through **solution action** (solution sinks) or may start as solution features. If the bottom of a sinkhole forms the roof of an underground void or cave, it might collapse, leaving a large hole opening into the cave or void below, known as **collapse sinks.** Solution sinks are more common than collapse sinks.
- Quite often, sinkholes are covered with a soil mantle and appear as shallow water pools. Anyone stepping over such pools might sink, similar to quicksand in deserts. The term **"doline"** is sometimes used to refer to collapse sinks.

Valley Sinks or Uvalas:

- When **sinkholes and dolines join together due to slumping of materials** along their margins or **roof collapse of caves, long, narrow to wide trenches** called **valley sinks or uvalas form.** Gradually, most of the limestone surface is eaten away by these pits and trenches, creating an extremely irregular surface with a maze of points, grooves, and ridges known as lapies.

Lapies and Limestone Pavements:

- Lapies are **ridges formed due to differential solution activity along parallel to sub-parallel joints.** These features can appear as grooves, flutes, or ridge-like formations in an open limestone field. Over time, these lapie fields may eventually transform into somewhat smooth limestone pavements.

Caves or Caverns:

- In areas with **alternating beds of rocks** (such as shales, sandstones, and quartzites) with limestones or dolomites in between, or in regions where limestones are dense, massive, and occur as thick beds, cave formation is prominent. **Water percolates down through the materials or through cracks and joints, moving horizontally along bedding planes.** It is along these bedding planes that limestone dissolves, resulting in long and narrow to wide gaps called **'Caves.'**
- Caves may form at different elevations depending on the limestone beds and intervening rocks, creating a maze of caves. Typically, caves have an **opening through which cave streams are discharged.** Caves with openings at both ends are referred to as **tunnels.**

Polje:

- A polje, also called a karst polje or karst field, is a **large, flat-bottomed depressions formed by the collapse of a large underground cavern or by the dissolution of surrounding rock.** These features are distinguished by their vertical side walls, **flat alluvial floors, independent drainage systems within the basin, irregular borders, and often a central lake.** Poljes are essentially large, closed basins with an elliptical shape, which can cover areas up to 258 square kilometers. They are commonly found in the karst regions of the former Yugoslavia and Jamaica.

Depositional Landforms by Groundwater:

Limestone caves are known for their fascinating depositional landforms, primarily formed from **calcium carbonate.** The key processes involved include the **dissolution** and **deposition** of calcium carbonate from carbonated water, which contains dissolved carbon dioxide.

Stalactites: ceil

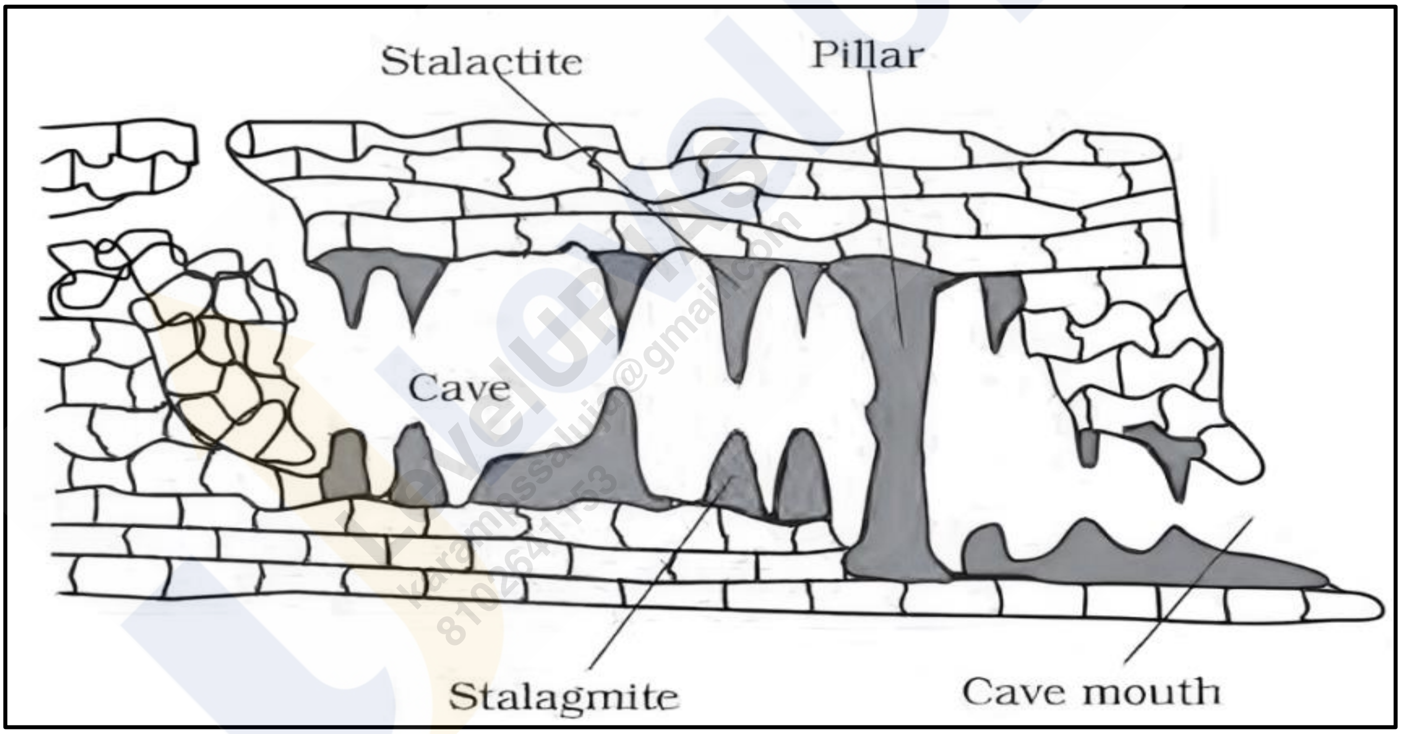
- Stalactites are **icicle-shaped** formations that **hang from the ceilings of caves.** They vary in diameter, generally being wider at their bases and tapering towards the ends. Stalactites form when **water dripping from the cave's ceiling carries dissolved calcium carbonate.** As the **water evaporates or loses its carbon dioxide, calcium carbonate is deposited,** creating these hanging formations.

Stalagmites: ground

- Stalagmites **rise from the cave floor directly beneath stalactites.** They develop from the dripping water that falls from the ceiling or from the thin pipe of the stalactite above. Stalagmites can take various shapes, including columns, discs, or rounded forms with smooth, bulging ends. They may also have a crater-like depression at the top.

Pillars:

- Pillars are formed when **stalactites and stalagmites grow together and merge, creating large, vertical columns.** These formations can vary widely in diameter and shape, reflecting the continuous deposition of minerals from both the ceiling and the floor.



Land form by G. Water

E

- → Sinkhole
- → Swallowhole
- → Dolin
- → Uvala
- → cave

D

- → Pillar
- → Stalactite
- → stalaqmite

Water table below limestone → KARST TOPOGR

Rock Joint → KARST TOPOGR

Aridd semi-arid region → KARST TOPOGR

Huge volume of thick bedded limestone → KARST TOPOGR

Lf Glacier

Erosion

(1) Tarn, cirque OR corre
(2) Ushaped valley
(3) Hanging
(4) Pyramidical peak
(5) Arete

D

- Eske
- Moraine
- Drumlin {egg shaped}
- outwash plain
- Till plain

Landforms Made by Glaciers:

Glaciers are **massive, moving bodies of ice,** often referred to as **"rivers of ice."** They form on land where climatic conditions support the accumulation of snow and ice. This occurs in regions where the seasonal snowfall exceeds the seasonal melting. Over time, **layers of snow accumulate, compact, and recrystallize into ice,** leading to the formation of glaciers.

Types of Glaciers:

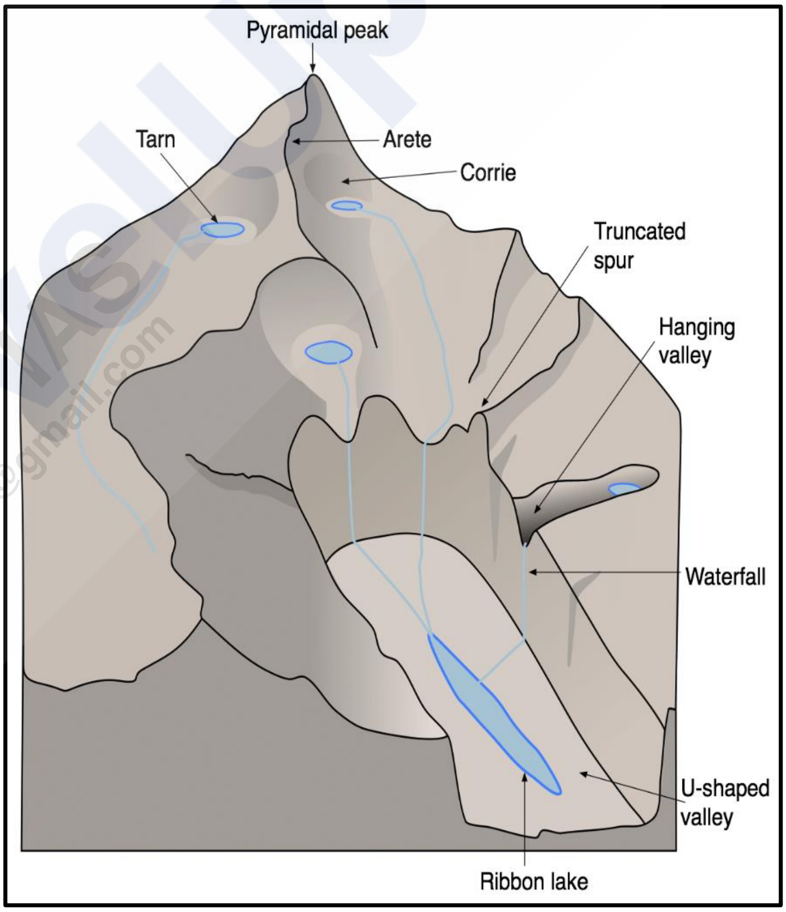
Continental Glaciers:

- Continental glaciers, also known as **ice sheets, are thick ice masses that cover vast areas of land.** They form in **polar regions** and can reach **thicknesses of several thousand meters.** Continental glaciers primarily build up at their centers and spread outward in all directions. **Examples - Greenland Ice Sheet, Antarctic Ice Sheet** and **Arctic Ice Sheets.**

Mountain or Valley Glaciers:

- Mountain or valley glaciers form in **high mountain regions and flow down existing valleys.** Their size and shape are influenced by the valley's slope and width. **Examples - Fedchenko Glacier, Siachen Glacier** and **Gangotri Glacier.**

Erosional Landforms by Glaciers:



Cirques or Corrie:

- Cirques are common landforms in glaciated mountains, often **found at the heads of glacial valleys.** They are **deep, long, and wide troughs or basins with steep, concave walls.** Cirques are formed by the erosion of ice as it moves down the mountainside. After the glacier melts, a **cirque lake, or tarn,** may form within the cirque. Sometimes, multiple cirques can be arranged in a stepped sequence.

Glacial Valleys or U-Shaped Valleys:

- Glacial valleys are **U-shaped troughs** with **broad floors and steep, smooth sides.** These valleys may contain debris or moraines and can have lakes formed by glacial meltwater. In some cases, **these glacial troughs can be inundated with sea water, forming fjords - deep, narrow sea inlets** typically found in high-latitude regions. Fjords have strikingly steep cliffs and are a dramatic example of glacial erosion.

Hanging Valleys:

- Hanging valleys are **formed by tributary glaciers that join larger glaciers.** These tributaries have less ice and, therefore, less erosive power, resulting in **valleys that are higher than the main glacier valley.** Waterfalls often form at the junction of hanging valleys and main valleys.

Horns and Serrated Ridges (Aretes):

- **Horns -** These are **sharp, pointed peaks** formed where **three or more cirques erode the mountain headward** until they converge. The intersecting walls create a steep, pointed peak.
- **Serrated Ridges (Aretes) -** These are **narrow, saw-toothed ridges** formed between cirques due to **progressive erosion.** They have a sharp crest and a zig-zag outline.

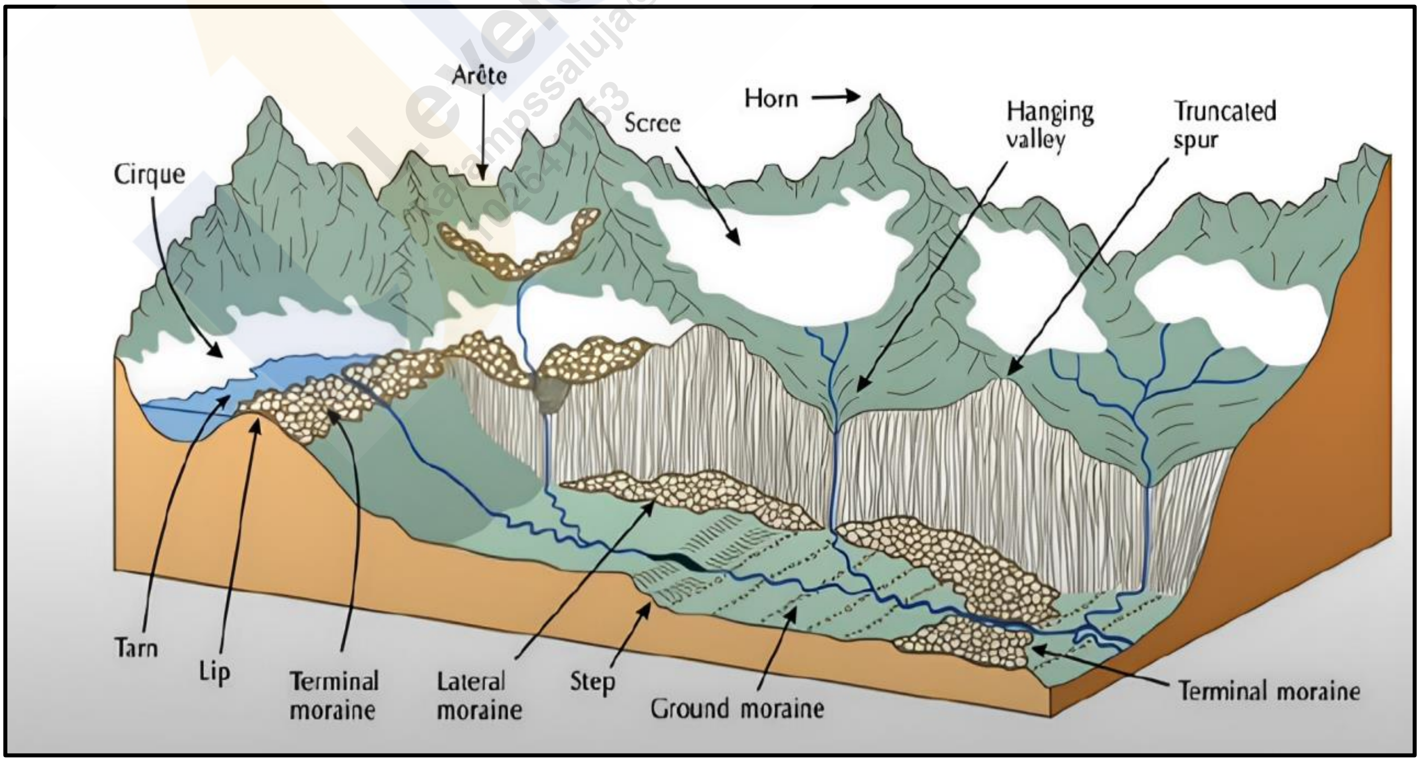
Depositional Landforms by Glaciers:

Glacial Till:

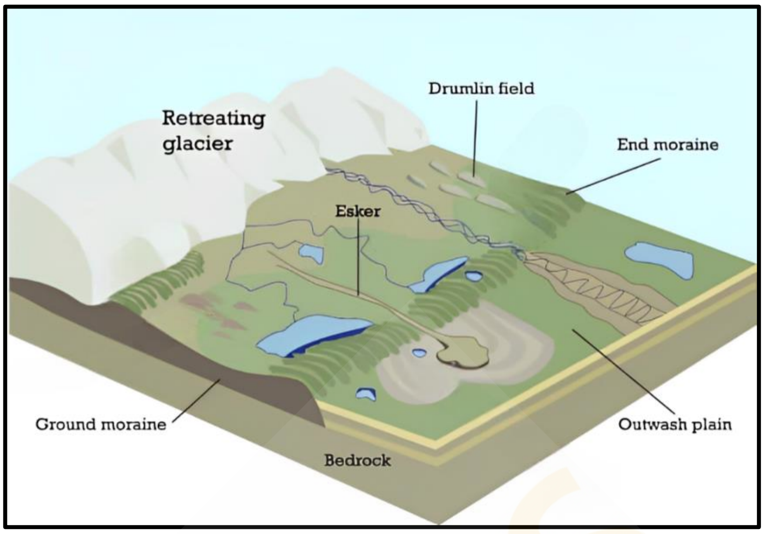
- Glacial till refers to the **unsorted and unstratified material deposited directly by melting glaciers.** This material includes a **mix of coarse and fine debris.** Some finer debris is transported by meltwater streams, which carry and deposit it further downstream. These deposits are known as outwash deposits, which are characterized by a more stratified and sorted structure compared to glacial till.

Moraines:

- Moraines are **elongated ridges formed from the accumulation of glacial till.** They are categorized based on their position relative to the glacier:
    - **Terminal Moraines -** These are ridges of **debris deposited at the furthest advance of the glacier,** marking its maximum extent.
    - **Lateral Moraines -** These ridges **run parallel to the sides of a glacier** and are formed from material accumulated along the glacier's edges.
    - **Ground Moraines -** When a **glacier retreats, it often leaves behind a sheet of till covering the valley floor,** known as a ground moraine. This deposit can be **irregular and uneven.**
    - **Medial Moraines -** Located in the **center of a glacial valley between two lateral moraines,** medial moraines are created from the merging of two lateral moraines. They may sometimes appear similar to ground moraines due to their less distinct formation.



Eskers:



- Eskers are **long, winding ridges composed of sand and gravel** that form from sediments **deposited by meltwater** flowing through tunnels or channels beneath or on top of a glacier. As the glacier melts, the sediment left behind creates these **long, narrow ridges.** These ridges often follow the direction of glacial flow and can extend for over 100 kilometers.

Drumlins:

- Drumlins are **elongated, oval-shaped hills formed from glacial drift beneath a glacier or ice sheet.** They are aligned in the direction of the glacier's flow and are typically low hills up to 1.5 kilometers in length and 60 meters in height. Drumlins are **steeper on the side facing the glacier's advance and taper off on the opposite side.** When found in clusters, this feature is known as '**basket of eggs' topography.**

Outwash Plains:

- Outwash plains are **flat areas composed of sediment washed out from terminal moraines** by meltwater streams. The meltwater sorts and redeposits materials such as sand and fine sediments, creating plains that are often **suitable for specialized agriculture** due to their sandy soils.

Kettle Lakes:

- Kettle lakes are depressions **formed in outwash plains when blocks of ice become buried in sediment and later melt.** These shallow, sediment-filled bodies of water are remnants of the glacial ice that once covered the area.

Landforms Made by Waves and Currents or Coastal Landforms:

Coastal processes are among the **most dynamic and often the most destructive forces** shaping our planet. Understanding these processes is crucial, as they can bring rapid and dramatic changes to coastal landscapes. Coastal changes can be both erosional and depositional, often varying with the seasons.

Most coastal changes are driven by the action of waves. When **waves break, they hurl water with tremendous force against the shore, causing a churning of sediments on the sea bottom.** This constant impact significantly alters coastlines. Storm waves and tsunamis can affect even more drastic changes in a shorter period compared to normal waves. As the wave environment fluctuates, so does the intensity of breaking waves.

Factors Influencing Coastal Landforms:

- **Configuration of Land and Sea Floor -** The physical **layout of the coast and the underwater topography** play a crucial role.
- **Coastal Movement -** Whether the coast is **advancing** (emerging) seaward or **retreating** (submerging) landward.

Assuming sea level remains constant, two primary types of coasts help explain the evolution of coastal landforms:

1. **High, rocky coasts** (submerged coasts).
2. **Low, smooth, and gently sloping sedimentary coasts** (emerged coasts).

High Rocky Coasts:

- High rocky coasts are characterized by **drowned river valleys and highly irregular shorelines, often indented with fjords.** These coastlines typically show **steep hillsides** that plunge sharply into the water, with **erosion features being dominant initially** and minimal depositional landforms present.
- Along high rocky coasts, **waves crash with great force, sculpting the hillsides into cliffs.** Over time, the relentless pounding of waves causes these cliffs to recede, leaving behind a **wave-cut platform** in front of the sea cliff. This process gradually smooths out the shoreline's irregularities.
- The materials eroded from the sea cliffs break into **smaller fragments, become rounded, and are deposited offshore.** With ongoing cliff erosion and material deposition, a wave-built terrace eventually forms in front of the wave-cut terrace. The eroded materials also provide a supply for longshore currents and waves, which deposit them as beaches along the **shore and as bars** (long ridges of sand and/or shingle parallel to the coast) in the nearshore zone. When these bars rise above water, they are **called barrier bars.** A barrier bar attached to the headland of a bay is known as a **spit.** When spits or barrier bars block the mouth of a bay, they create a **lagoon.** Over time, these lagoons can fill with sediments from the land, forming **coastal plains.**

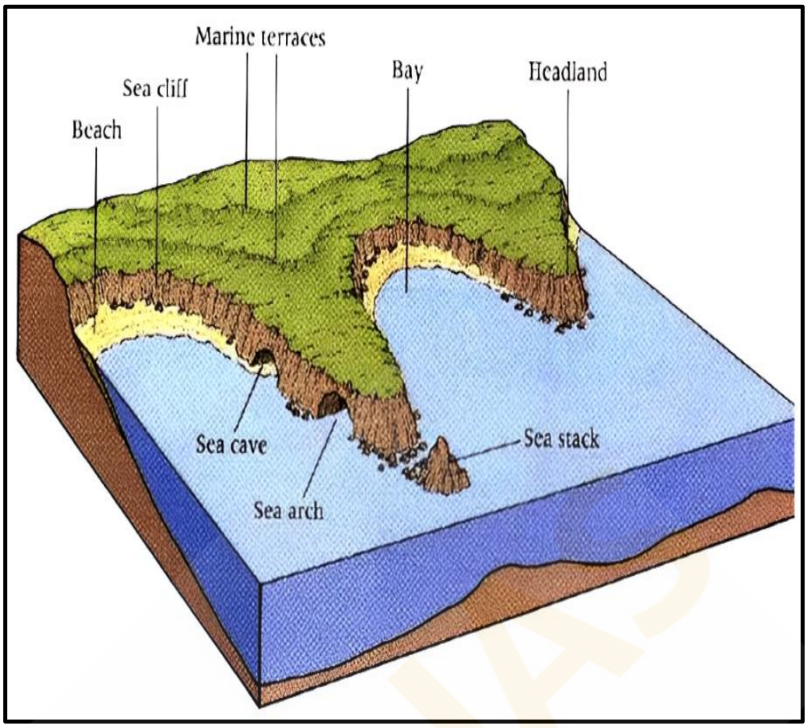
Low Sedimentary Coasts:

- Low sedimentary coasts typically feature **rivers extending their lengths by building coastal plains and deltas.** These coastlines appear smooth, with **occasional incursions of water in the form of lagoons and tidal creeks.** The land slopes gently into the water, and marshes and swamps are common.
- On gently sloping sedimentary coasts, breaking waves churn the bottom sediments, moving them to **build bars, barrier bars, spits, and lagoons.** Over time, lagoons can transform into swamps and eventually into coastal plains. The stability of these depositional features relies on a steady supply of sediments. However, storm waves and tsunamis can cause significant changes regardless of sediment supply. Large rivers that deliver substantial amounts of sediments form deltas along these low sedimentary coasts.

Do you know?

*The **west coast** of our country is **a high rocky retreating coast.** Erosional forms dominate in the west coast. The **east coast** of India is a **low sedimentary coast.** Depositional forms dominate in the east coast.*

Erosional Coastal Landforms:



Sea Cliffs:

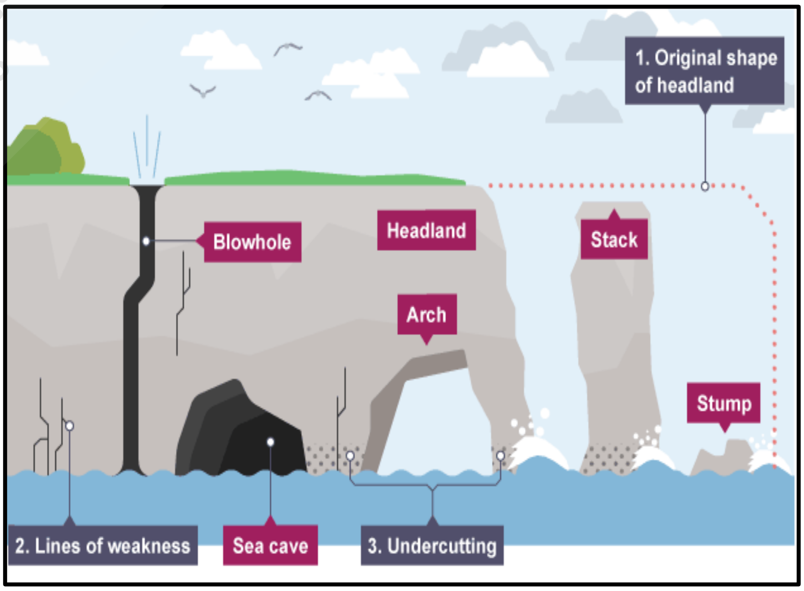
- Sea cliffs are prominent features formed by marine erosion. These **steep, rocky coastlines rise almost vertically above sea level.** The steepness of a cliff is determined by variations in geological structure, lithology, and the relative rates of subaerial weathering and erosion. The **lower part of a cliff experiences the maximum impact from sea waves,** especially if it is composed of softer rocks like limestone, which erode more rapidly than the harder rocks typically found in the upper part. Over time, the **unsupported upper sections project outward and eventually collapse into the sea,** leaving behind a vertical wall known as a **cliff.** Numerous sea cliffs can be observed along the Konkan coast of India.

Sea Caves:

- Sea caves form when waves crash against headlands, **eroding weaker sections such as cracks.** Differential erosion creates a hollow in the **lower part of the rock, especially if it consists of softer material, while the upper part remains harder.** The continuous pounding of waves compresses air inside the hollow, which expands when the waves recede, subjecting the rocks to significant pressure. **Over time, this process enlarges the hollow into a sea cave.** The cave can further expand and deepen, especially if the roof collapses due to the lack of support.

Blowholes:

- Blowholes develop **when sea caves grow larger and waves begin to crash into the back wall of the cave, forcing air and water upwards through the roof.** This process creates a vertical opening, or blowhole, in the headland. Blowholes are rare and require a vertical crack line to be effectively exploited by the erosive forces of waves.



Sea Stacks:

- Sea stacks are **isolated pillars of rock that form when a sea arch collapses.** Initially, sea arches develop from the merging of sea caves that form on opposite sides of a headland. When these arches give way, the remaining rock columns are known as sea stacks.

Sea Terraces:

- Sea terraces are **rock platforms that form when a sea cliff, along with its wave-cut platform, is uplifted above the sea level.** These terraces represent former positions of sea level and are often evident as **flat, bench-like landforms** above the current shoreline.

Depositional Coastal Landforms:

Beaches:

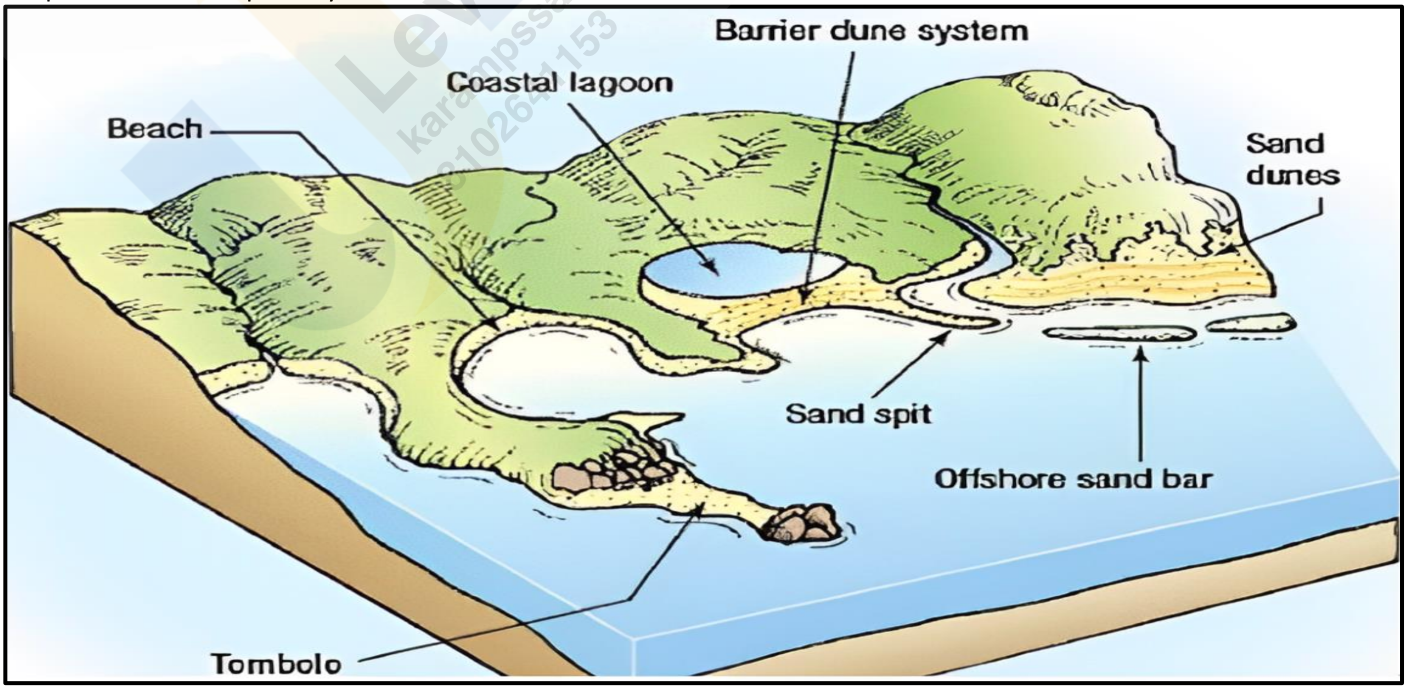
- Beaches serve as **natural barriers** between waves and upland features, such as dunes. They are **narrow, gently sloping patches of land** found along oceans, seas, rivers, and lakes. Composed of **unconsolidated sand, gravel, rocks,** and **seashells,** beaches are constantly reshaped by waves and wind.
- **Calmer weather** and gentle, long-period waves, known as swells, **tend to deposit sediment onshore,** building up the beach. In contrast, **steep waves from storms and strong winds erode the beach by transporting sediment offshore.** Wind blowing across the beach moves finer particles, which accumulate to form dunes when they encounter barriers such as vegetation.
- Notable examples of beaches include Radhanagar Beach in the Andaman and Nicobar Islands, Agonda Beach in Goa, Marina Beach in Chennai, Seminyak Beach in Bali, and Santa Monica Beach in California.

Spits and Bars:

- **Spits -** A sand spit is a **linear deposit of sediment** that extends from the land into the open water. Spits typically form where the **coastline bends inward from the prevailing direction of longshore drift.** The sediment accumulates in the direction of the updrift coast, creating a **narrow, elongated landform.**
- **Bars -** A sandbar, also known as an **offshore bar,** is a **submerged or partially exposed ridge of sand built by wave action offshore from a beach.** These bars are formed by the **deposition of sand** and other sediments, creating a barrier that can be seen during low tide or through the clear waters of shallow coastal areas.

Sand Dunes:

- Sand Dunes are **mounds or ridges of sand deposited by the wind** immediately landward of the beach. **Primary dunes, closest to the beach, are usually lightly vegetated** with grasses, while **secondary dunes farther inland may have denser vegetation,** including shrubs and trees.
- **Vegetation traps and holds windblown sand, stabilizing the dunes.** Large primary dunes provide protection from flooding and wave action for upland areas and can replenish eroded beaches. Secondary dunes offer additional protection if the primary dune is breached.



Tombolos:

- A **sandy isthmus connecting a small island to the mainland.** It is created by the deposition of sediments by waves and tides, which gradually build up to connect the two land masses.

Lagoons:

- Lagoons are **shallow, often brackish water bodies located between barrier bars or spits and the mainland,** formed when sandbars or spits enclose an area of water behind them.

Estuaries:

- Estuaries are **partially enclosed coastal bodies of water,** where freshwater from rivers and streams mixes with seawater. The deposition of sediments from river runoff creates unique habitats.

Mudflats:

- **Soft, muddy sediments** exposed during low tide.

Salt Marshes:

- Coastal wetlands dominated by **salt-tolerant vegetation,** formed by the deposition of fine sediments carried by tidal waters.

Landform Made by Wind or Desert Landforms:

Wind acts as a **geomorphic agent across all terrestrial environments,** with its effects being **most pronounced** in arid regions (Deserts) characterized by fine-textured soils and sediments and sparse or absent vegetation. **A desert is a landscape characterized by extremely low precipitation, leading to harsh conditions for both plant and animal life.** The scarcity of vegetation leaves the ground surface vulnerable to denudation processes. **Approximately one-third of the Earth's land surface is classified as arid or semi-arid,** encompassing regions such as the Polar Regions, which, due to minimal precipitation, are often referred to as polar or **"cold deserts."**

Deserts can be categorized based on various factors including the amount of precipitation, temperature, causes of desertification, or geographical location. For instance, about one-fifth of the world's land is desert. Some deserts are so devoid of life that they are termed **"true deserts."**

Types of Deserts:

- **Hamada (Rocky Desert) -** These deserts are characterized by **extensive areas of bare rock, stripped of sand and dust by wind action.** The exposed rocks are often smooth and polished, resulting in a **highly sterile environment.**

- **Reg (Stony Desert) -** Reg deserts feature large expanses of **angular pebbles and gravels** that are **not easily moved** by wind. These deserts are generally more accessible than sandy deserts and can **support large herds of camels.**

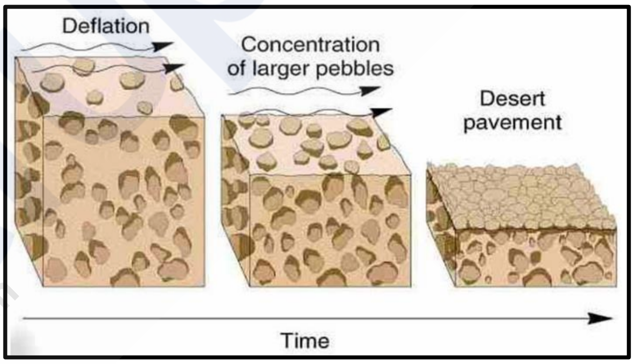
- **Erg (Sandy Desert) -** Also known as **"seas of sand,"** these deserts are dominated by vast stretches of **sand dunes** formed by wind deposition. The dunes often shift with the wind, creating a **dynamic landscape.**

- **Badlands -** Characterized by **gully and ravine formations** on slopes and rock surfaces due to **water erosion,** badlands are **unsuitable for agriculture** and often lead to the abandonment of the region by its inhabitants.

- **Mountain Deserts -** These deserts are found in **highland regions,** such as plateaus and mountain ranges, where erosion has sculpted the landscape into **rugged peaks and uneven terrain.** The steep slopes are often marked by **dry valleys known as wadis,** which have irregular edges shaped by **frost action.**

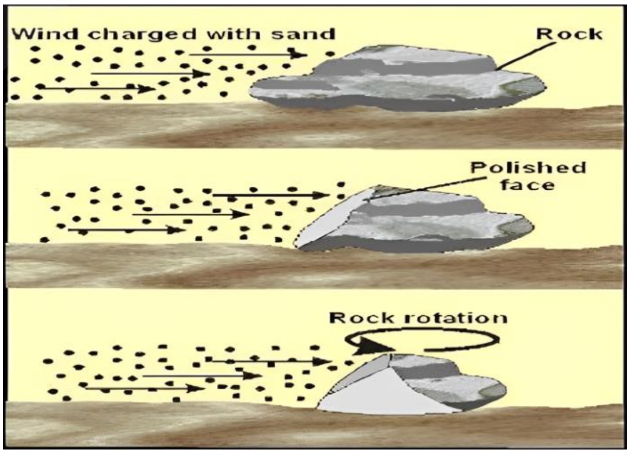
Action of Wind:

Deflation:



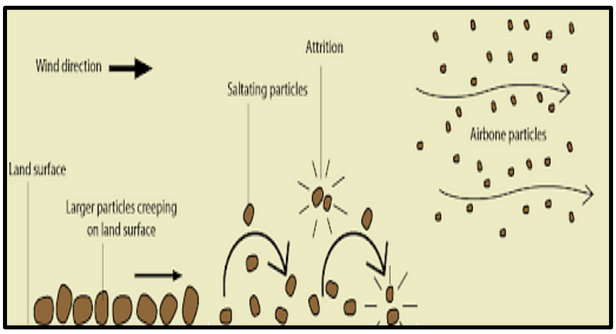
- Deflation involves the **removal of loose surface materials by the wind,** which lifts and carries away these particles. The efficiency of deflation depends on the wind's ability to transport different sizes of material. **Fine dust and sand can be transported over long distances and may even be deposited beyond desert margins.** Over time, deflation leads to the formation of large depressions known as deflation hollows.

Abrasion:



- Abrasion occurs when **wind-driven sand particles strike rock surfaces, effectively sandblasting them.** This process results in the **scratching, polishing, and wearing away of the rocks.** Abrasion is most intense near the base of rocks, where the wind carries a higher concentration of particles. For this reason, **desert telegraph poles often have metal coverings up to several feet above the ground** to protect them from wind erosion.

Attrition:



- Attrition happens when **wind-borne particles collide with each other during transport.** These collisions cause the **particles to wear down, resulting in smaller, rounded grains** known as **millet seed sand.**

Erosional Landforms by Wind:

Rock Pedestals (Mushroom Rocks):

- Rock pedestals, also known as **mushroom rocks,** are formed by the **sandblasting action of wind against projecting rock masses.** As wind-driven sand **erodes the softer layers of rock,** it leaves behind irregular edges and a distinctive shape. Over time, this process carves grooves and hollows into the rock, resulting in grotesque, pillar-like formations. The **erosion is most intense near the base** where friction is greatest, leading to the development of mushroom-shaped rocks with a broader top and a narrower base.

Mesas and Buttes:

- A mesa, from the Spanish word for **'table,'** is a **flat-topped** landform with **steep sides.** It features a resistant horizontal top layer that protects the underlying rock from erosion by both wind and water. Mesas can be found in canyon regions, such as Arizona, or on fault blocks like Table Mountain in South Africa. Over time, continued erosion can reduce mesas to **isolated flat-topped hills called buttes**, which are characterized by their reduced size and steep sides.

Zeugen:

- Zeugen are **tabular landforms where a hard, resistant rock layer covers a softer underlying layer.** The differential erosion between these layers creates a ridge-and-furrow landscape. Initially, mechanical weathering opens joints in the surface rocks, and wind abrasion further erodes the softer layer, forming deep furrows. The **harder rock stands above these furrows as ridges.** Zeugen can range from 10 to 100 feet in height, and ongoing wind erosion gradually lowers these features while widening the furrows.

Yardangs:

- Yardangs **resemble zeugen but differ in their orientation.** In yardangs, **the hard and soft rock layers are arranged vertically.** Winds, aligned with the direction of the prevailing **winds, erode the softer rock into narrow corridors, leaving behind steep-sided ridges of harder rock.** These long, narrow landforms, called yardangs, are characterized by their distinct, streamlined shapes formed by continuous wind abrasion.

Inselbergs (Island Mountains):

- **Inselbergs, or island mountains,** are isolated **residual hills** that **rise sharply** from the surrounding terrain. Typically composed of **granite or gneiss,** they feature very steep slopes and **rounded tops.** These landforms are remnants of a once extensive plateau that has been largely eroded away.

Pediments and Pediplains:

- **Pediments** are **gently sloping, rocky surfaces** found at the base of **mountains in desert regions.** These landforms develop through the erosion of the mountain front, primarily due to **lateral erosion by streams and sheet flooding.** Initially, erosion begins along the steep margins or sides of tectonically controlled features. As pediments form, they typically start with a steep wash slope and a cliff or free face above. Over time, the wash slope and free face retreat backward, a process known as **parallel retreat of slopes or backwasting**. This retreat extends the pediment backward at the expense of the mountain front, eventually reducing the mountain to a low, featureless plain known as a pediplain.

Playas:

- In **desert basins surrounded by mountains and hills,** the drainage often directs toward the center of the basin, where sediment gradually accumulates to form a nearly level plain. During periods of sufficient moisture, this plain may be covered by a **shallow water body.** These shallow lakes are termed playas and typically **retain water only temporarily due to high evaporation rates.** Playas often **accumulate salts,** and when these salts cover the playa plain, it is referred to as an alkali flat.

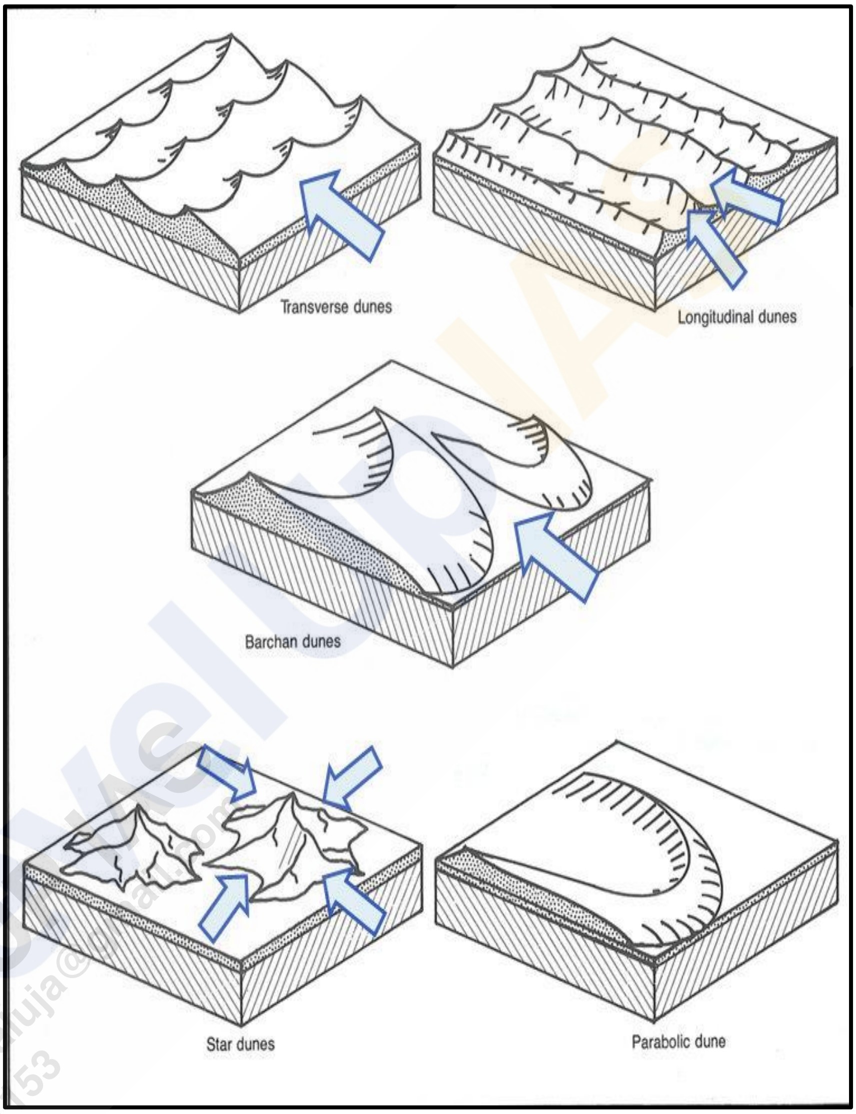
Deflation Hollows and Caves:

- **Deflation Hollows -** Also known as blowouts, these are depressions formed by the removal of loose particles from the ground by wind. Typically small in scale, blowouts can occasionally expand to several kilometers in diameter.
- **Caves -** As wind-driven sand and other particles continually impact rock faces within deflation hollows, these depressions can deepen and widen, evolving into cave-like structures.

Depositional Landforms by Wind:

Dunes:

- Dunes are hills of sand formed by the accumulation and movement of sand shaped by the wind. They can be classified as either active (constantly moving) or fixed (stabilized by vegetation). Dunes are most prominent in erg deserts, where vast seas of sand are continuously reshaped by wind.
- **Types of Dunes**:
  - **Barchans** - These **crescent-shaped dunes** have their **points or wings oriented away from the wind direction.** They form in areas where wind direction is consistent and moderate, and the sand moves across a relatively uniform surface.
  - **Parabolic Dunes** - Resembling **reversed barchans,** parabolic dunes develop on partially vegetated sandy surfaces. The **wind direction remains the same** as for barchans, but the **presence of vegetation influences their shape.**
  - **Seif Dunes** - Seif dunes resemble barchans but have **only one wing or point.** They form in areas where wind conditions shift, causing the single wing of a seif to extend significantly in length and height.
  - **Longitudinal Dunes** - These dunes appear as **long ridges aligned with the direction of prevailing winds.** They form where the sand supply is limited and wind direction remains constant. Longitudinal dunes are typically elongated and low in height.
  - **Transverse Dunes** - Aligned **perpendicular to the wind direction,** transverse dunes form when there is a consistent wind direction and a sand source that extends at right angles to the wind. These dunes can be very long but generally low in height.



Ripples:

- Ripples are small, regular, wavelike undulations that form **perpendicular to the direction of the prevailing wind.** These features are created as wind moves across loose sediment, arranging it into distinct, patterned shapes.

Loess:

- Loess is a type of wind-deposited sediment consisting of fine, **fertile dust.** It is **yellowish, friable, and highly porous, allowing water to sink in quickly, which often results in a dry surface.** Loess can form extensive deposits, as seen in the **Loess Plateau of northwest China,** which covers about 250,000 square miles and reaches depths of 200 to 500 feet. Known locally as **'Huangtu'** (yellow earth), loess deposits in China are similar to those in Germany, France, Belgium, and parts of the Midwestern USA. In these regions, loess was deposited from material at the edges of ice sheets during the Ice Ages and is sometimes called adobe.

===== XXXXX =====

(1) Wind
(2) River
(3) underground Water
(4) Wave
(5) Glacier

Shape Landform ← Geomorphic Process ← (1)–(5)

Geomorphic Process → Erosion, Deposition

Land form by River

Erosion
(1) Canyon
(2) Gorge } V-shaped
(3) Pothole
(4) Plunge Pool & Waterfall

Deposition
1. Delta
2. Flood Plain
3. Alluvial Plain
4. Riverline Island
   ↳ Majuli in Brahmaputra
5. Natural levee

Special landform
Meander & oxbow lake
2. River Terrace